

BioSpecInfo

Traceability Matrix — BioSpecInfo

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Traceability Matrix — BioSpecInfo

Field	Value
Software	BioSpecInfo
Author	Samuele Pio Provenzano
Version described	bsi-v186
Purpose	Connect every declared requirement to the implementation that realises it and to the bench that verifies it.

How to read this matrix. Every row is a closed chain: a requirement, the point in the code that realises it, and the bench that verifies it works. A row without a bench is an **unverified** requirement and is declared as such, not hidden: the usefulness of a matrix lies precisely in making the gaps visible.

The outcomes of the benches cited here are reported in full, with their complete output, in [../evidence/RAPPORTO-VERIFICA.md](#), which is generated automatically.

1. Scientific requirements

ID	Requirement	Implementation	Verification bench
SCI-01	Structural data for drugs must be consistent with the declared molecular weight	FARM_DATA in <code>index.html</code>	<code>tools/verifica-farmaci.js</code> — RDKit ↔ literature comparison over 233 entries
SCI-02	Functional groups must be recognised on the structure, not on the text of the SMILES	<code>bsi-spettri.js</code> §1, SMARTS patterns via RDKit	<code>test_spettri</code> — 8 reference molecules, additive recognition
SCI-03	A molecule with several functional groups must show them all	<code>bandeDaGruppi()</code> , additive construction	<code>test_spettri</code> §1 — acetylsalicylic acid: ester and acid
SCI-04	Spectra must respect the conventions of their technique	<code>svgIR()</code> , <code>makeNMRsvg()</code> , <code>drawUVSpectrum()</code> , <code>drawMSSpectrum()</code>	<code>test_assi</code> (SVG), <code>test_assi_canvas</code> (canvas)
SCI-05	A computed spectrum must be reproducible	every <code>Math.random()</code> removed from the traces	<code>test_assi</code> §3, <code>test_spettri</code> §8 — byte-for-byte comparison
SCI-06	The height of ¹ H peaks must represent the integration, or not be asserted at all	<code>makeNMRsvg()</code> ; <code>_nmrSvg()</code> draws bands of uniform height	<code>test_assi</code> §4

ID	Requirement	Implementation	Verification bench
SCI-07	Conjugation must act on the individual carbonyl	SMARTS <code>acidoConiug</code> , <code>estereConiug</code> , <code>chetoneConiug</code> , <code>aldeideConiug</code>	<code>test_spettri</code> §1 — aspirin 1760/1690; acetophenone 1690
SCI-08	An ambiguous physical constant must not be resolved arbitrarily	<code>costante_fisica</code> in <code>bsi-ai-hub.js</code> — collects every candidate	<code>test_costanti</code> — 45 checks, including the refusals for ambiguity
SCI-09	The limits of the predictors must be declared to the user	chart headers; <code>docs/06</code> §3.5	<code>verifica_guida</code> — consistency between promises and code
SCI-10	Molecular symmetry must determine the selection rules correctly	<code>ssimm</code> section in <code>index.html</code>	<code>test_simmetria</code> — 26 checks: normal modes, centre of inversion, mutual exclusion
SCI-12	A chromatogram must follow from the parameters, not be drawn by hand	<code>scroma</code> section: $\sigma = t_{R}/\sqrt{N}$, R_s from distance and widths	<code>audit_stabilita</code> (opening and drawing), inspection of the values against the definitions
SCI-13	A calibration line must declare the uncertainty of its coefficients and recognise the two ways it misleads	<code>staratura</code> section: ordinary least squares, F-test line $\leftrightarrow$ parabola, deleted residual	Verified on the three example sets included: linear ($F = 0.3$), non-linear ($F = 186.6 > 7.71$), with an outlier (1 detected)
SCI-14	A QSAR model must be evaluated on scaffolds never seen, not on analogues from the same set	<code>bsi-cheminfo.js</code> §9 — <code>divisionePerScaffold()</code> alongside <code>divisioneCasuale()</code>	<code>test_cheminfo</code> — the scaffold split leaves no scaffold shared between training and test
SCI-15	A model score must be compared against chance, not presented on its own	<code>bsi-cheminfo.js</code> §9 — retraining on scrambled labels, eight repetitions	<code>test_cheminfo</code> — verified in both directions : on a learnable signal R^2 0.394 beats all eight twins (margin 0.287); on random labels R^2 -0.199 does not beat them
SCI-16	Two spellings of the same molecule must not be counted twice, nor land on opposite sides of the split	<code>bsi-cheminfo.js</code> §1 — deduplication on the canonical SMILES after the largest fragment	<code>test_cheminfo</code> — <code>OC(=O)C</code> and <code>CC(=O)O</code> collapse into one entry; salts lose the counterion
SCI-17	Similarity between molecules must be computed, not estimated	<code>bsi-cheminfo.js</code> §3 — Tanimoto and Dice over Morgan, RDKit, pattern and MACCS fingerprints	<code>test_cheminfo</code> — compared against hand-computable values, including the $0/0 = 1$ convention
SCI-18	A validation on a single split, over a small set, is a noisy number: it must come with its dispersion	<code>bsi-cheminfo.js</code> §9-bis — <code>validazioneIncrociata()</code> over scaffold-grouped folds	<code>test_cheminfo</code> — the folds cover every molecule once, no scaffold sits in two folds , and the same seed gives the same folds

ID	Requirement	Implementation	Verification bench
SCI-19	A result that changes on every run is verifiable by nobody	seeded mulberry32 generator for splits, scrambles and folds; the seed is written into the report	test_cheminfo — two calls with the same seed give identical folds; the method report declares the seed
SCI-20	Results must be able to leave the tool, and the method must leave with them	bsi-cheminfo.js \$9-ter — esportaCsv() and rapportoMetodo(), both formed in memory in the browser	test_cheminfo — columns consistent with the header, a name containing a comma quoted, residual computed, empty cell for missing activity; the report carries seed, fingerprint, verdict, limits
SCI-21	A number presented as structural similarity must come from a fingerprint, and one implementation must compute it	rdkit_lab.html loads bsi-cheminfo.js and uses its tanimoto() and fingerprint(); the eight-bit similarity has been removed	test_cheminfo — page and engine give the same Tanimoto over all 28 pairs , deviation exactly 0 ; the fake-similarity functions no longer exist; every reference structure is valid and carries the monoisotopic mass of the drug it names
SCI-22	Conversions between molecular languages must be checked against an external source or against hand calculation, and coordinates written to a file must be real and declared	bsi-molingue.js — seventeen outputs , from SMILES to CML	test_lingue — 7 literature InChI keys , full SMILES → molfile → SMILES round-trip, 26 hand-written names, 7 required refusals ; mass composition against hand calculation (alanine C 40.44%, aspirin C 60.00%); XYZ and PDB with non-null coordinates and declared 2D
SCI-23	Angles, lengths and dihedrals are measured on the coordinates, and on flat coordinates they are not shown	dot product for angles, cross product for dihedrals	test_mol3d — tetrahedron 109.4712° , water 104.47°, CO ₂ 180°, BF ₃ 120°, ammonia 106.13°; dihedrals 0°, 60°, 90°, 180°; ethanol C-C 1.509 Å; and the refusal tested in both directions

2. AI agent requirements

ID	Requirement	Implementation	Verification bench
AI-01	A model retired by the provider must not block the assistant	modelloSuggestitoDaErrore(), bsi_modelli_ko blacklist	test_404 — 36 checks
AI-02	A provider unreachable from the browser must be remembered and demoted	bsi_prov_ko, 24 h TTL, providerUtilizzabili()	test_ko (35), browser_ko (37)
AI-03	A temporary overload must produce a retry, not an error	erroreTemporaneo(), TENTATIVI_TEMPORANEO	test_503 — 40 checks

ID	Requirement	Implementation	Verification bench
AI-04	An imposed wait longer than 10 s must switch provider if a ready one exists	ATTESA_TROPPO_LUNGA_MS	test_attesalunga — 14 checks
AI-05	The provider's token ceiling must be learned from the error, not hard-coded	tettoDaErrore(), MARGINE_TETTO	test_tetto — 32 checks
AI-06	Opaque reasoning signatures must be returned to the provider	capture and replay in appendAgentTurn()	test_firma — 31 checks
AI-07	Two overlapping submissions must not corrupt the history	_turnoInCorso in send()	test_doppioinvio — 7 checks
AI-08	The user must be able to measure which providers they can actually reach	provaTuttiIFornitori()	browser_prova — 29 checks
AI-09	Paid providers must not be offered as free ones	separation by PROVIDERS[id].free	browser_prova \$D2
AI-10	A documented route must exist to remove CORS and rate limits	proxy/spectra-proxy.js + in-app guide	test_guidaproxy (12), browser_proxy (15), browser_proxyui (17)

3. Stability requirements

ID	Requirement	Implementation	Verification bench
STA-01	A long session must accumulate neither DOM nodes nor timers	section lifecycle management	audit_stabilita \$1 — 91 sections × 5 rounds
STA-02	Exhausting localStorage must not make the app unusable	guarded writes; persistent warning in the File Manager	audit_quota (10), test_filemanager (15)
STA-03	No rejected promise may remain unhandled	systematic .catch()	audit_promesse — 22 checks, healthy network and dead network
STA-04	Corrupted saved data must not prevent startup	loadJSON() with fallback	audit_stabilita \$4
STA-05	A slow network must not be treated as an absent one	ATTESA_RETE_MS = 3500 in sw.js	test_sw \$3 and \$3-bis — measured 3,507 ms
STA-06	An update must not destroy work in progress	BSI_SW_UPDATED message, decision taken page-side	test_sw \$4
STA-07	Browsing molecules must not exhaust the WebGL contexts	reuse of the 3D viewer at all 5 creation points	test_visore3d — from 7 constructions down to 1

ID	Requirement	Implementation	Verification bench
STA-08	An unavailable IndexedDB store must be declared, not fatal	dbStore() over a promise; persistent warning	test_filemanager §2

4. Interface and accessibility requirements

ID	Requirement	Implementation	Verification bench
UI-01	Every page must open without JavaScript errors	—	audit_stabilita §5 — 14 pages
UI-02	Charts must be sharp on high-density screens	bsiNitido() on the canvas contexts	audit_grafici — 40 canvases
UI-03	The app must work at 390 px width	auto-fit grids, no fixed column	audit_mobile — the document's scrollWidth across all 91 sections at 390 px; audit_stabilita for errors in a phone viewport
UI-04	The user must be able to know which version they are running and force the update	"Updates" entry in the ✨ panel	test_aggiorna — 9 checks
UI-05	Data deletion must be selective and reversible in its choices	bsiCancellaDati() by group	browser_reset — 24 checks
UI-08	Surfaces that appear only after an action must also meet WCAG AA contrast	the six panels of the Cheminformatics section	test_cheminfo — the analysis is run, the panels opened in turn, 460 text elements measured, 0 defects , with the WCAG formulas rewritten inside the bench
UI-09	Opening a section must not block the page	3Dmol's first render() moved out of the click; the 296 synthesis figures drawn in 8 ms slices	test_fluidita — 91 sections traversed , median switch 14 ms, worst 162 ms, 0 sections above 100 ms ; plus the opposite proofs: the WebGL canvas appears anyway, all 296 figures exist shortly after, and printing does not come out mute
UI-10	The interface must switch among many languages, declaring how much each is translated, and carrying the writing direction with it	bsi-lingue.js with fourteen languages , a searchable selector, dir="rtl" for Arabic	test_lingue — 170 skeleton elements out of 170 in each of the thirteen languages , coverage computed by applying the dictionary and counting; writing direction tested in both directions; navigation categories and the search placeholder included
UI-11	The molecule must show how it holds together, and the viewer must choose the angle	bsi-mol3d.js: formation from dust with staggered arrival, and atom selection by click	test_mol3d — the canvas pixels at 130 ms and 2.3 s, the arrival order (skeleton before hydrogens) measured on each atom's progress, and the selection of 1, 2, 3 and 4 atoms

5. Security and privacy requirements

ID	Requirement	Implementation	Verification bench
SEC-01	No API key may be present in the repository	keys only in <code>localStorage</code> or in the Worker's secrets	<code>tools/verifica-sicurezza.js</code> — 295 tracked files, 8 credential shapes
SEC-02	No personal data may leave the device without an explicit action	local-first architecture, telemetry disabled	<code>audit_rete</code> — direct verification : a canary value seeded into 71 stores of the user's data, the application used across 91 sections on 6 pages, and the URL, headers and body of every request inspected. Plus <code>verifica-sicurezza</code> on the two exit mechanisms
SEC-03	Passwords must not appear in clear text in the source	SHA-256 in <code>file_manager.html</code>	<code>tools/verifica-sicurezza.js</code>
SEC-07	No credential may EVER have entered the repository, not even in a commit later fixed	no key has ever been committed; keys live in <code>localStorage</code> or in the Worker's secrets	<code>audit_storia</code> — 1,551 distinct versions of text files across 390 commits , 9 patterns tested in both directions; fails on a shallow clone, because it would measure less surface
SEC-04	No conflict marker may reach publication	—	<code>verifica_guida</code> §12 — 58 text files

SEC-03 — declared limitation. GitHub Pages serves static files: any page-side access control can be bypassed by whoever reads the source. The File Manager's protection is a deterrent, **not** a security control, and the documentation says so. Furthermore the password remained in clear text in the git history until its removal, and taking a secret out of the files does not take it out of the history: `git log -p` hands it to anyone. The only effective remedy was to change it, and that **has been done** at version `bsi-v186`. The old one remains in the history and no longer opens anything.

6. Requirements not yet verified automatically

Declared for completeness. They are covered by documentary inspection or by manual testing, and their automation is planned.

ID	Requirement	Current coverage
UI-06	Full WCAG 2.1 AA conformance	the mechanical part is automated across 13 pages and 91 sections (<code>tools/verifica-accessibilita.js</code> , 33,642 text elements: it was 19,751 before text over gradients entered the measurement). Of the 1,069 contrast defects that emerged, none remain: 0 measured over those same 91 sections, and the value is recorded as a baseline the bench defends. Unlabelled fields are zero : see the box in <code>docs/09</code> §4. What remains outside is text inside SVGs and text over a real background image, counted on every run (D-09)
PERF-01	First-paint time on a low-end device	manual cross-device testing (<code>docs/02</code> §4)
SCI-11	Structures of 2 entries that are not single molecules (it was 6)	not representable : Ivermectin is a mixture of homologues, Coartem a combination of two active ingredients. The other four were closed by taking the structure from ChEMBL, see <code>docs/06</code> §2.4
PERF-02	Code coverage of the verification benches	partially measured : 49.89 % of statements over the widest path (<code>audit_copertura</code>). Whole-battery coverage and branch coverage remain unmeasured; see §8
UI-07	Behaviour on Firefox and WebKit	not verified — the battery runs on Chromium only; see §8

7. Overall coverage

Category	Requirements	Verified by a bench	Coverage
Scientific (SCI-01...23)	22	22	100 %
AI agent (AI-01...10)	10	10	100 %
Stability (STA-01...08)	8	8	100 %
Interface (UI-01...11)	9	9	100 %
Security (SEC-01...07)	5	5	100 %
Automated total	54	54	100 %
Not automated (§6)	5	0	0 %
Declared total	59	54	91 %

Coverage is computed over the requirements **declared in this document** and must not be confused with code coverage: it measures how many requirements have a bench verifying them, not how many lines are executed during the tests.

Why two totals. Reporting only the 100 % of automated requirements would be true and misleading at once: it is 100 % of what was *chosen* to be automated. The number that matters to an assessor is the second one — **54 requirements verified out of 59 declared** — and the five that are missing are listed by name in §6, not summarised into a percentage.

This table is checked by `tools/verifica-documenti.js`, which counts the identifiers actually present in the document and fails if the totals written here do not match. It is a check born of a real error: the totals said 36 over 9 scientific categories while the identifiers defined were 10, and `SCI-10` appeared twice — once as a verified requirement and once as an unverified one.

8. Declared coverage gaps

These are not defects: they are things that have **not been measured**, and that therefore cannot be asserted.

Gap	What it entails	Why it is so
Partial code coverage	Measured: 49.89 % of statements over the widest path a bench walks. It is not the coverage of the whole battery, and it is statements, not branches	Chromium's profiler collects it inside the engine, with no build and without rewriting the source: the obstacle belonged to the tool (<code>c8</code> , <code>istanbul</code>), not to the problem. Value recorded; <code>audit_copertura</code> fails if it falls
Chromium only	Behaviour on Firefox and WebKit is verified by hand, not by a bench	The battery uses <code>playwright-core</code> , which downloads a single engine. In the verification environment the CDN for the other engines answers 403 to the network policy: they cannot be installed there
No visual regression	An unintended graphical change would not be caught	Spectra are deterministic and comparable byte for byte (<code>SCI-05</code>): the comparison exists on the traces, not on the whole page
2 entries without a structure	Two entries out of 178 have no structure to show: Ivermectin is a mixture of homologues, Coartem a combination of two active ingredients. It was six	See <code>06-Scientific-Accuracy-Data-Provenance.md</code> §2.4
Performance on slow devices	First-paint time is not measured on low-end hardware	It requires physical devices; the test is manual

Document updated to version `bsi-v186`.